



Replicating Procedures

1. Working with a partner, assemble ten blocks into an unusual structure. Write directions that others can use to replicate that structure without seeing it.
2. Exchange directions with another team. Replicate the team's structure by following their directions.
3. Compare each replicated structure to the original. Identify which parts of the directions were clear and accurate, and which were unclear or misleading.

ANALYZE AND CONCLUDE

1. **Evaluate** How could you have written better directions?

2. **Construct an Explanation** Based on what you have learned in this investigation, explain why it is important that scientists write procedures that can be replicated.



Figure 1-7

Mangrove Swamp

In tropical areas, mangrove swamps serve as the ecological equivalents of temperate salt marshes. The results of the salt marsh experiment suggest that nitrogen might be a limiting nutrient for mangroves and other plants in these similar habitats.

Peer Review Scientists share findings with other scientists by publishing their hypotheses, experimental methods, results, and analysis in scientific journals. Papers submitted to these journals are reviewed by anonymous, independent reviewers who work in the same field. Reviewers look carefully for mistakes, oversights, unfair influences, or fraud, in techniques and analysis. Their goal is to ensure that articles meet the highest standards of quality. This process is called peer review. ***Publishing articles in peer-reviewed scientific journals allows researchers to share ideas. It also allows other scientists to evaluate and test the data and analysis.*** Peer review does not guarantee that a piece of work is correct, but it does certify that the work meets standards set by the scientific community.

Sharing Knowledge and New Ideas After research has been published, it enters the dynamic marketplace of scientific ideas. How do new data fit into existing scientific understanding? For example, the finding that growth of salt marsh grasses is limited by available nitrogen sparks more questions: Is the growth of other plants in the same habitat also limited by nitrogen? What about the growth of different plants in similar environments, such as the mangrove swamp shown in **Figure 1-7**? These logical and important questions leads to new hypotheses that must be independently supported or rejected by controlled experiments.



READING CHECK Explain Why is it necessary for other scientists to evaluate the findings of scientists in their field?